

EXERCISE 1 SQL FUNDAMENTALS

QUESTION 1: Selecting and Sorting

1. SELECT *
FROM employees;

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. SELECT DISTINCT department
FROM employees;

department
IT
HR
Finance
Marketing

3. SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;

Eva
Alice
Cathy
David
Brian

3. SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

QUESTION 2: Filtering with WHERE

5. SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. SELECT id, first-name, last-name, department, salary
 FROM employees
 WHERE department = 'finance'
 AND salary > 60000

id	First-name	last-name	department	salary
3	Cathy	Zulu	Finance	65000

7. SELECT id, first-name, last-name, department
 FROM employees
 WHERE department = 'HR'
 OR department = 'Marketing';

id	First-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. SELECT id, first-name, last-name, department
 FROM employees
 WHERE department != 'IT';

id	First-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. SELECT id, first-name, last-name, department
 FROM employees
 WHERE department IN ('IT', 'HR', 'finance');

id	First-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT id, first-name, last-name, department, salary, city
 FROM employees
 WHERE department = 'IT'
 AND salary > 65000
 AND city = 'Johannesburg';

id	first-name	last-name	department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg